

DATA TYPES

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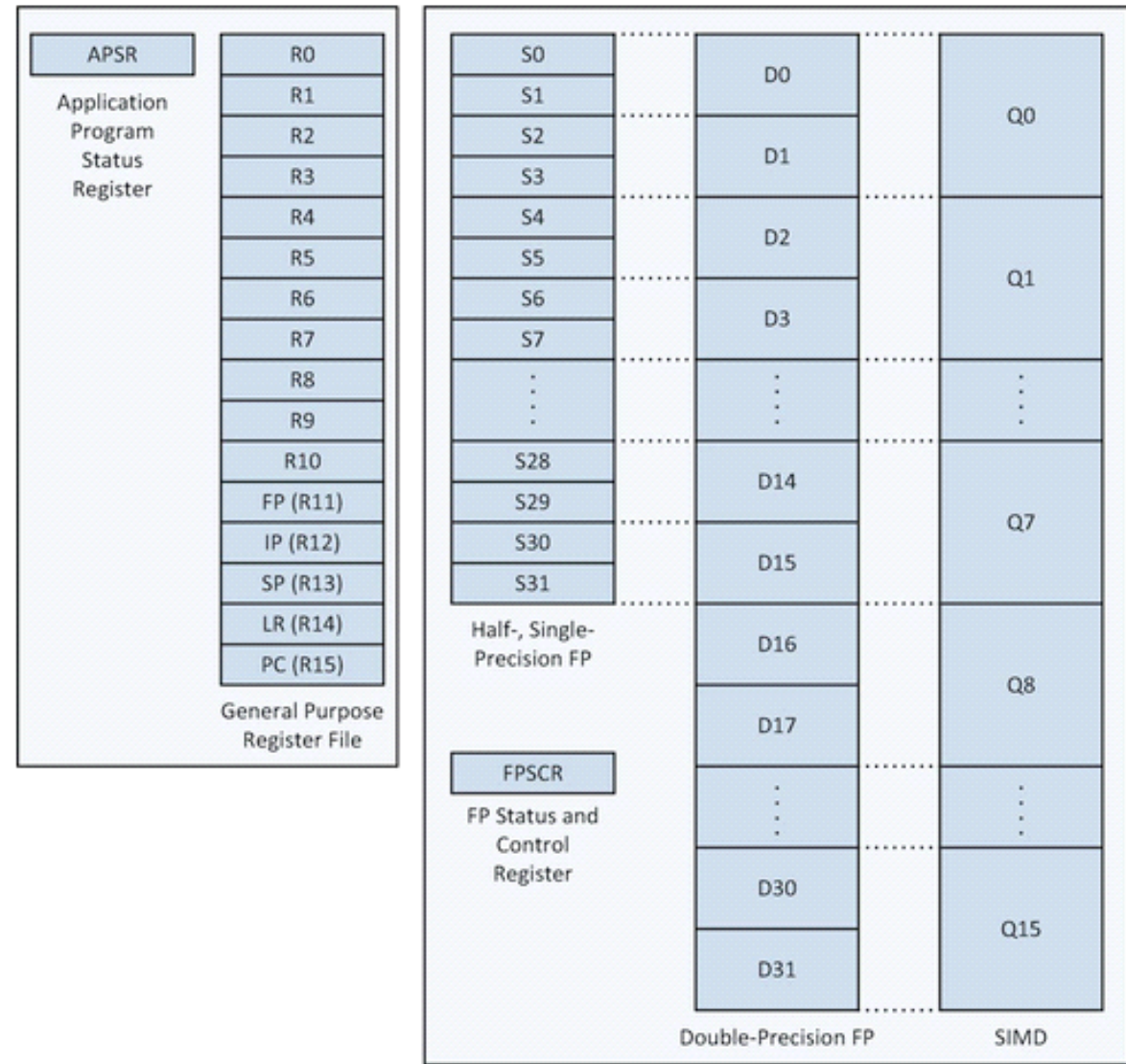
Type	Size (bits)	C++ Type	<stdint>
Signed integers	8	char	int8_t
	16	short	int16_t
	32	int, long	int32_t
	64	long long	int64_t
Unsigned integers	8	unsigned char	uint8_t
	16	unsigned short	uint16_t
	32	unsigned int, unsigned long	uint32_t
	64	unsigned long long	uint64_t
Floating-point	16	n/a	n/a
	32	float	n/a
	64	double	n/a

able 1-2: AArch32 numerical data types

REGISTERS +flagy

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AArch32 registre



APSR - Application Program Status Register
teda príznačkový register

General-Purpose Registers

- R0-R10 - pracovné registre
- FP (R11) - Frame-Pointer (obdobu BP) lokálne premenne prociek
- IP (R12) - intra-procedure-call scratch register - niečo ako segmentový register (niečo pre 32-bit address space) **veneers**
- SP (R13) - stack pointer
- LR (R14) - link register -suvisi s navratom z volanej procedury.
- PC (R15) - Programm Counter - tiež môže byť použitý ako obyčajný register, ale iba na zdrojovej strane, ako cieľ nie

FPSCR - Floating-Point Status and Control Register
príznačkový register numerického koprocera

S0-S31 - Single precision (Float?)

D0-D31 - Double Precision (Double?)

Q0-Q15 - ~~Quad Precision~~/SIMD registre

ASPR - teda priznakovy register ma bity s nasledujucim vyznamom:

Bit Position	Symbol	Name
31	N	Negative condition flag
30	Z	Zero condition flag
29	C	Carry condition flag
28	V	Overflow condition flag
27	Q	Overflow or saturation flag
26:24		Reserved 0
23:20		Reserved for system features or future expansion
19:16	GE[3:0]	Parallel add/sub greater than or equal flags
15:0		Reserved for system features or future expansion

Table 1-3: APSR status bit fields

NZCV - najdolezitejsie flagy ... aj sa spolocne oznacuju pri popisoch ako "NZCV"

V - je vysledok prilis maly, alebo prilis velky v signed matematike, 127+4 v 8-bit napriklad

Q - pouziva pri jednoduchych SIMD operaciach, aj ked miesto neho existuje rozsireny SIMD register.

Registre S0-S31, D0-D31, Q0-Q15 ... su prekryvne podla obrazku ...cize nejde o paralelne/nezavisle registre.

SIMD operacie sa daju robit ako s celymi cislami, tak s floatingom.

INSTRUCTION samples

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Type	Example Instruction	Analogous C++ Statement
Immediate	mov r0,#42	r0 = 42
	add r1,r0,#8	r1 = r0 + 8
	add r0,#17	r0 += 17
	lsl r2,r1,#2	r2 = r1 << 2
Register	mov r1,r0	r1 = r0
	and r2,r1,r0	r2 = r1 & r0
	mul r2,r1,r0	r2 = r1 * r0
	smull r4,r5,r0,r1	r4:r5 = r0 * r1
Memory	ldr r0,[sp]	r0 = *sp
	str r7,[r4]	*r4 = r7

Table 1-4: Examples of A32 instruction operands

mul r4,r5,r0,r1 (r5:r4)64bit = (r0)32bit * (r1)32bit unsigned operacia
smull r4,r5,r0,r1 (r5:r4)64bit = (r0)32bit * (r1)32bit signed operacia

Addressing operandov

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- offset addressing register +/-konstanta
- pre-indexed addressing
aktualizuje sa bazova adresa, teda index, este pred vykonanim pamatovej operacie
- post-indexed addressing
aktualizuje sa bazova adresa, teda index az po vykonani pamatovej operacie
(nieco ako stringove operacie x86 stosb, lodsrb)
- PC relative addressing

Mode	Example	Analogous C++ Statements
Offset	ldr r2,[r1]	r2 = *r1
	ldr r2,[r1,#8]	r2 = *(r1 + 8)
	ldr r2,[r1,r0]	r2 = *(r1 + r0)
	ldr r2,[r1,r0,ls1 #2]	r2 = *(r1 + (r0 << 2))
Pre-indexed	ldr r2,[r1,#8]!	r2 = *(r1 + 8); r1 += 8
	ldr r2,[r1,r0]!	r2 = *(r1 + r0); r1 += r0
	ldr r2,[r1,r0,ls1 #2]!	r2 = *(r1 + (r0 << 2)); r1 += r0 << 2
Post-indexed	ldr r2,[r1],#8	r2 = *r1; r1 += 8
	ldr r2,[r1],r0	r2 = *r1; r1 += r0
	ldr r2,[r1],r0,ls1 #2	r2 = *r1; r1 += r0 << 2
PC relative	ldr r2,label	r2 = *(pc + label_offset)

able 1-5: Examples of A32 memory addressing modes

Vysvetlivka jedneho maleho tajomstva:

Ako sa ARM CPU dostane k datam na opacnom konci pamate s 32-bit adresou ??,
ak adresy operandov skrz dlzku instrukcie presne 32-bitov
nikdy nemozu dosiahnut dlzku 32-bitov ??

.... mno ... najprv si CPU cusne adresu vzdialenej premennej ako konstantu do registra sposobom [PC+maly_offset]
... a potom **ldr Rx,[Ry]**
konstanta s adresou premennej je mozno aj z pamatoveho segmentu v ktorom je kod ...aby konstanta s adresou
premennej bola blizsko instrukcie [PC+signed(8-bit)??].